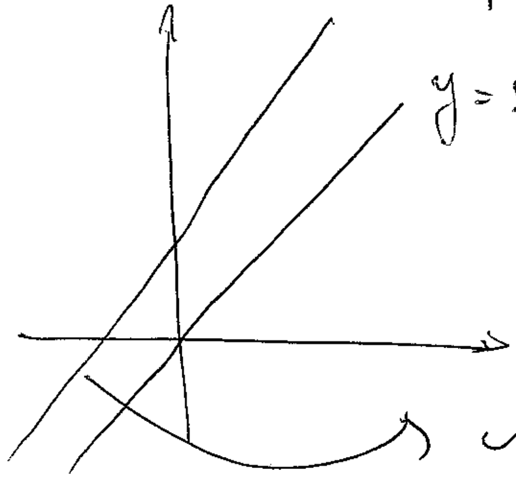
Section 2-8 Affine Spaces



$y = x$ is a subspace of $\mathbb{R}^2$

$y = x+1$ is not a subspace of $\mathbb{R}^2$ → not closed under scalar multiplication and addition

$(1,0)$ is on $y = x+1$

$(2,0) = 2(1,0)$ is not

Def Let V be a vector space, $x_0 \in V$, $U \subseteq V$ is a subspace, then the set

$L = x_0 + U = \{ x_0 + u : u \in U \}$ is called affine subspace of V.

Ex: $y = x+1$ is an affine subspace of $y = x$

- Affine subspace does not contain the origin if $x_0 \notin U$.
- Affine subspace $= U$ if $x_0 \in U$.
- Affine subspace can be described by parameters

$$x = x_0 + \lambda_1 b_1 + \dots + \lambda_k b_k$$

with $\lambda_1, \dots, \lambda_k \in \mathbb{R}$ where $\{b_1, \dots, b_k\}$ is a basis of U.

- for one-dimensional space, affine subspaces are called lines and can be written as $x = x_0 + \lambda b$ where x_0 is a point in the line and b is the direction.
- In two-dimensions they are called planes

$$x = x_0 + \lambda_1 b_1 + \lambda_2 b_2$$

"support point" plane

- $\{x \mid Ax = b\}$ is an affine subspace

$= \{$Any solution of $Ax = b\} + \{x \mid Ax = 0\}$